

When Averaged Field Regularity Fails to Rank Few-Step Generative Paths

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Generative ODEs and Flow Matching

A continuous normalizing flow is the ODE

$$\frac{dX_t}{dt} = v_\theta(t, X_t), \quad X_0 \sim p_0,$$

with $v_\theta : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$. The input is time and the current state; the output is an instantaneous velocity. A generating field u_t and density p_t obey the continuity equation

$$\partial_t p_t + \nabla \cdot (p_t u_t) = 0.$$

Flow Matching

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, X_t \sim p_t} \left\| v_\theta(t, X_t) - u_t(X_t) \right\|^2.$$

Once a velocity field is learned, generation requires numerically integrating the ODE. [Lipman et al., 2023]

Conditional Flow Matching and stochastic interpolation

A two-sided interpolant with independent endpoints is

$$X_t = \alpha(t)X_0 + \sigma(t)X_1, \quad X_0 \perp X_1,$$

with pathwise derivative $\dot{X}_t = \dot{\alpha}(t)X_0 + \dot{\sigma}(t)X_1$. The marginal velocity is $b_t(x) = \mathbb{E}[\dot{X}_t \mid X_t = x]$. This is the same generating field as on the previous slide: $b_t = u_t$.

Conditional Flow Matching

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, q(x_1), p_t(x|x_1)} \left\| v_\theta(t, x) - u_t(x \mid x_1) \right\|^2.$$

Lipman Theorem 2 then gives $\nabla_\theta \mathcal{L}_{\text{CFM}} = \nabla_\theta \mathcal{L}_{\text{FM}}$.

Training uses accessible conditional velocities. Averaging over compatible endpoint pairs recovers the marginal field that transports p_t . In the Gaussian laboratory below, that field is known exactly: $b(t, x) = A(t)x + c(t)$. [Lipman et al., 2023, Albergo

et al., 2025]

The few-step question

$$(\alpha, \sigma) \longrightarrow p_t \longrightarrow b_t \longrightarrow \text{solver stages} \longrightarrow \hat{p}_1.$$

NFE is the number of evaluations of the velocity field, not the number of solver steps. At budget B : Euler uses B steps, Heun uses $B/2$ steps, RK4 uses $B/4$ steps.

Two paths with the same endpoints can induce different fields and different errors under the same solver:

$$\text{linear: } (\alpha, \sigma) = (1 - t, t), \quad \text{VP: } \left(\cos \frac{\pi t}{2}, \sin \frac{\pi t}{2} \right).$$

With independent endpoint sampling, the linear interpolant uses the product coupling, not an optimal transport coupling. Trigonometric VP is not Lipman VP diffusion path (eq. 18).

Question

Can an averaged regularity score rank probability paths by their fixed-NFE endpoint error?

[Lipman et al., 2023, Albergo et al., 2025, Liu et al., 2023, Chen et al., 2026, Sabour et al., 2024]

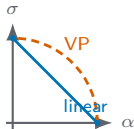
Exact Gaussian benchmark

Independent Gaussians $X_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$, $X_1 \sim \mathcal{N}(\mu_1, \Sigma_1)$:

$$m_t = \alpha \mu_0 + \sigma \mu_1, \quad Q(t) = \alpha^2 \Sigma_0 + \sigma^2 \Sigma_1, \quad C_t = \dot{\alpha} \alpha \Sigma_0 + \dot{\sigma} \sigma \Sigma_1.$$

$$b(t, x) = A(t)x + c(t), \quad A(t) = C_t Q(t)^{-1}, \quad c(t) = \dot{m}_t - A(t)m_t,$$

with $A(t) \in \mathbb{R}^{d \times d}$ and $c(t), x, b(t, x) \in \mathbb{R}^d$.



Coefficient plane (schematic).

[Albergo et al., 2025, Lipman et al., 2023, Gelbrich, 1990]

A controlled Gaussian laboratory isolates discretization error:

- exact marginal field
- no neural-network training error
- numerical endpoint remains Gaussian
- Gaussian W_2 is available in closed form
- solver discretization is isolated

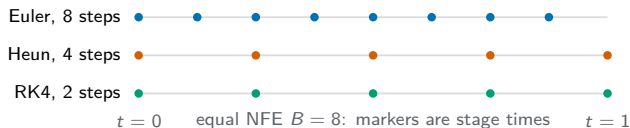
Candidate regularity criterion

Chen, Vanden-Eijnden, and Xu use averaged squared Jacobian Lipschitzness. For the affine field, $\nabla_x b(t, x) = A(t)$ is state-independent, so

$$\mathcal{R}[b] = \int_0^1 \mathbb{E}[\|\nabla_x b(t, X_t)\|_2^2] dt = \int_0^1 \|A(t)\|_2^2 dt.$$

Here $\|\cdot\|_2$ is the spectral operator 2-norm, as implemented.

This quantity is a plausible stability or error-control proxy. Controlling or upper-bounding an error is weaker than correctly ordering two paths at a fixed small NFE. $\mathcal{R}$ averages over $[0, 1]$; a solver samples finitely many stages.



[Chen et al., 2026]

Frozen experimental protocol

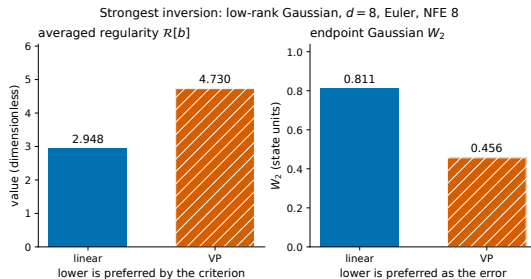
Each comparison holds target, solver, and NFE fixed, then compares linear versus VP. Source $N(0, I)$. Equal-NFE accounting throughout.

factor	frozen grid
targets	anisotropic (condition 4), seeded low-rank
dimensions	$d \in \{2, 8\}$
paths	linear vs trigonometric VP
solvers	Euler, Heun, RK4
NFE	8, 16, 32
configurations	72
paired comparisons	36
endpoint metric	Gelbrich W_2
precision	80-digit audit of all 72 values

14 ranking inversions among 36 prespecified paired comparisons in the frozen grid. Data:

`phase4_gaussian_reproduction_2026-07-24-v1:results.`

Main result



Experimental data. Strongest inversion: low-rank Gaussian, $d = 8$, Euler, NFE 8, metric W_2 . Linear $\mathcal{R} = 2.9476523251$, $W_2 = 0.8108540111$; VP $\mathcal{R} = 4.7295206355$, $W_2 = 0.4564779075$; margin 0.3543761036 (linear W_2 is 78% larger). Lower is preferred on both axes.

Result

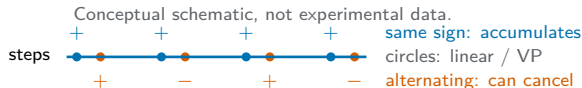
In this controlled commuting Gaussian benchmark, lower averaged regularity does not necessarily imply lower fixed-NFE endpoint error.

Mechanism: signed and transported defects

Centered commuting mode $x'_i = a_i(t)x_i$. Exact one-step factor $e_{i,n} = \sqrt{q_i(t_{n+1})/q_i(t_n)}$. A Runge-Kutta step multiplies by $r_{i,n}$. With S steps,

$$\prod_{n<S} r_{i,n} - \prod_{n<S} e_{i,n} = \sum_{n<S} (r_{i,n} - e_{i,n}) \left(\prod_{j<n} r_{i,j} \right) \left(\prod_{j>n} e_{i,j} \right).$$

Earlier factors are numerical, later factors are exact. No smallness assumption. Each local defect $r_{i,n} - e_{i,n}$ is signed; later dynamics transport it; defects can cancel; endpoint error depends on stage locations. An unsigned time average of $\|A\|_2^2$ does not encode those cancellations.



Grid-aware Euler construction

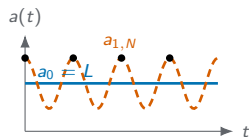
Fix $L > 0$ and integer $N \geq 1$, $h = 1/N$, left-endpoint Euler nodes $t_n = n/N$. Set $\epsilon_N = N(e^{L/N} - 1) - L > 0$ and $a_0(t) = L$, $a_{1,N}(t) = L + \epsilon_N \cos(2\pi Nt)$.

$$\int_0^1 a_0 = \int_0^1 a_{1,N} = L, \quad \text{exact endpoints } e^L.$$

$$\int_0^1 a_{1,N}^2 = L^2 + \frac{1}{2}\epsilon_N^2 > L^2 = \int_0^1 a_0^2.$$

At every node, $\cos(2\pi Nt_n) = 1$, so $1 + a_{1,N}(t_n)/N = e^{L/N}$ and $\prod_{n=0}^{N-1} (1 + a_{1,N}(t_n)/N) = e^L$. For the constant field, $(1 + L/N)^N < e^L$.

On this frozen grid, the higher-regularity field is integrated exactly, while the lower-regularity constant field is not. The field $a_{1,N}$ is chosen after the grid N .



Schematic, $N = 4$ shown. Not experimental data.

Scope and limitations

What the study establishes

- exact affine Gaussian benchmark
- frozen-grid empirical inversions
- signed transported-defect identity
- explicit grid-aware Euler non-implication

Current limitations

- commuting Gaussian covariances
- affine exact fields, not learned fields
- two main path families
- a finite solver and NFE grid
- Euler construction is grid-aware
- identity-style ODE parameterization
- exponential integrators are exact here
- no direct theorem for CTMC samplers

Next experiments: non-commuting covariances, learned fields, adaptive or optimized schedules, non-affine systems, and discrete rate-based samplers. [Lu et al., 2022a, Zhang and Chen, 2023]

Conclusion

- 1 Averaged field regularity may be useful for control or bounds, but it is insufficient by itself to rank fixed-NFE sampling paths in the tested benchmark.
- 2 Few-step endpoint error is governed by signed local defects, solver stage locations, propagation, and cancellation.
- 3 The controlled Gaussian model provides an exact diagnostic environment and an explicit non-implication example.

Research direction

For discrete generative models, the analogous question is how a probability path, its time-dependent rates, and a finite-step CTMC solver interact under a fixed computational budget.

A1. Equal-NFE accounting

NFE counts evaluate calls. Bookkeeping is not NFE.

$$n_{\text{steps}} = B / \text{evals per step}, \quad B \bmod \text{evals per step} = 0.$$

solver	evals / step	steps at $B = 8$	steps at $B = 32$
Euler	1	8	32
Heun	2	4	16
RK4	4	2	8

A higher-order solver receives fewer, more informative steps at the same field-evaluation budget. Equal step count would not be an equal-compute comparison.

A2. Affine Gaussian field

$X_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$, $X_1 \sim \mathcal{N}(\mu_1, \Sigma_1)$ independent in $\mathbb{R}^d$. Interpolant $X_t = \alpha X_0 + \sigma X_1$. Joint-Gaussian conditioning gives

$$b_t(x) = \dot{m}_t + C_t Q(t)^{-1}(x - m_t),$$

hence $A(t) = C_t Q(t)^{-1}$ and $c(t) = \dot{m}_t - A(t)m_t$. Then $b(t, \cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $A(t) \in \mathbb{R}^{d \times d}$, $c(t) \in \mathbb{R}^d$. Centered with $\Sigma_0 = I$: $A = \frac{1}{2} Q' Q^{-1}$, $c = 0$. This is the stochastic-interpolant marginal ODE. Linear $(1 - t, t)$ is independent coupling. It is not automatically a global OT coupling. [Albergo et al., 2025, Gelbrich, 1990]

A3. Gelbrich W_2 and $|r_i|$

For Gaussians,

$$W_2^2 = \|\hat{m} - \mu_1\|_2^2 + \text{tr}\left(\hat{C} + \Sigma_1 - 2(\Sigma_1^{1/2} \hat{C} \Sigma_1^{1/2})^{1/2}\right).$$

Reported values are $W_2 = \sqrt{W_2^2}$, not W_2^2 . In the centered commuting case, with numerical modal factors r_i ,

$$W_2^2 = \sum_{i=1}^d (|r_i| - \sqrt{\lambda_i})^2.$$

The absolute value is required if a coarse Euler factor is negative. [Gelbrich, 1990]

A4. Local defects, exact minus method

For $x' = a(t)x$ with smooth a , the leading coefficients of the local defect (exact factor minus method factor) are

$$\delta_{\text{Euler}} = \frac{1}{2}(a' + a^2)h^2 + O(h^3), \quad \delta_{\text{Heun}} = \left(-\frac{1}{12}a'' + \frac{1}{6}a^3\right)h^3 + O(h^4).$$

Euler and Heun sample different derivatives and different stages. RK4 is compared through its exact scalar stage polynomial against $\sqrt{q_i(t_{n+1})/q_i(t_n)}$. No memorized non-autonomous RK4 leading formula is used.

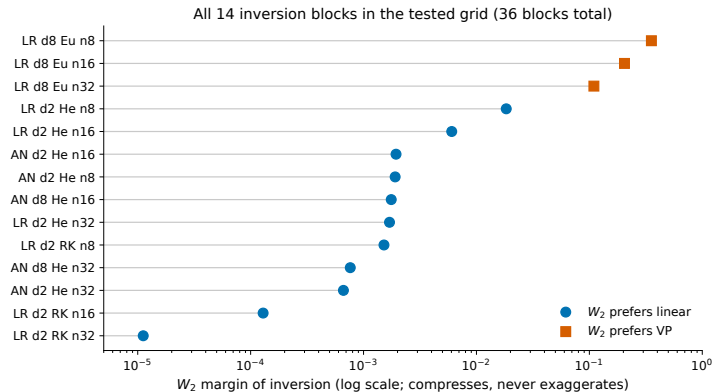
A5. Numerical precision

All 72 float64 W_2 values versus an 80-digit reference:

- maximum absolute difference $9.7050430488 \times 10^{-10}$
- smallest inversion margin $1.1188920612 \times 10^{-5}$
- ratio greater than 11,500

Source: phase4_precision_2026-07-24-v1. Mixture evidence was excluded after estimator calibration failure.

A6. All 14 inversion margins



Experimental data. Frozen Phase 4 grid. Logarithmic axis compresses, never exaggerates, magnitudes.

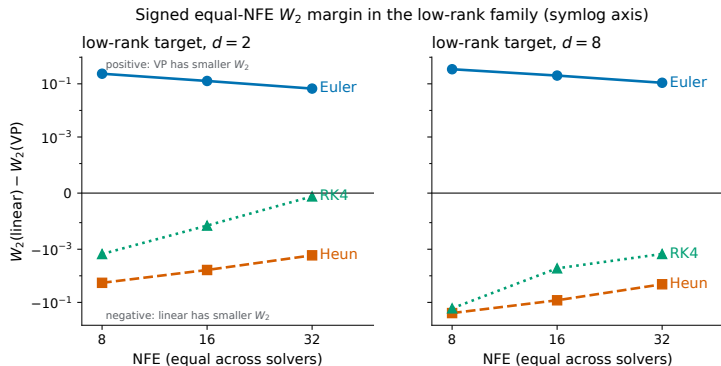
A7. Non-centered replication

Pre-registered family: source $\mathcal{N}(\mu_0, I)$ with $\mu_{0,i} = 0.75(-1)^i$, target $\mathcal{N}(\mu_1, D)$ with geometric spectrum of condition number 6, so $c(t) \neq 0$.

- 11 of 18 blocks invert
- $\mathcal{R}$ prefers VP (margin 0.7513) because $\mathcal{R}$ ignores $c(t)$ by definition
- W_2 prefers linear in those inverted blocks
- all 11 pass the 80-digit audit

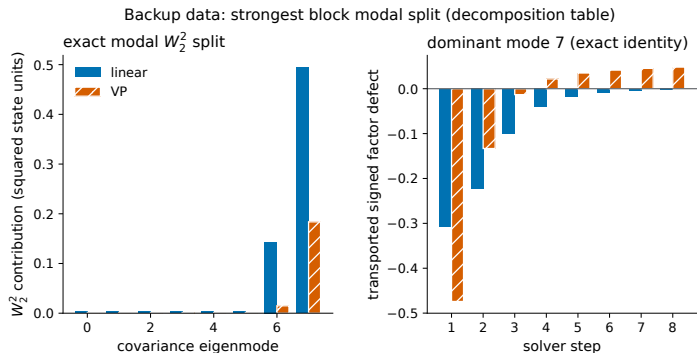
Still Gaussian, diagonal, and commuting. Supporting replication, not out-of-distribution evidence.

A8. Solver-path interaction



Experimental data. Positive: VP has smaller W_2 . Negative: linear has smaller W_2 . Low-rank family only. Equal NFE. Source: frozen Phase 4 results.

A9. Strongest-block modal split



Experimental data. Uses W_2^2 modal contributions, not W_2 . Decomposition table is post-hoc relative to the Phase 3 gate. Dominant mode (index 7) accounts for the bulk of the linear error.

A10. Averaged regularity proposal

Definition 3.2 of Chen, Vanden-Eijnden, and Xu is $A_2 = \int_0^1 \mathbb{E}[\|\nabla b_t(l_t)\|_2^2] dt$, spectral 2-norm. The paper also gives a transfer formula that optimizes a schedule relative to a linear reference, typically under a constraint such as $\alpha^2 + \sigma^2 = 1$. This talk ranks two fixed paths at the identity time parameterization. Reparameterization-based schedule optimization is a complementary use that the result does not address. [Chen et al., 2026]

A11. Exponential integrators

DPM-Solver, DPM-Solver++, and DEIS integrate a linear component exactly and approximate the remainder. On the present affine Gaussian fields those methods are exact, so the ranking question is vacuous for them. Learned fields are not affine. The limitation study therefore concerns generic fixed-stage Runge-Kutta discretization, which remains relevant when the field is nonlinear. [Lu et al., 2022a,b, Zhang and Chen, 2023]

A12. CTMC analogy

In a continuous-time Markov chain the generator $Q_t(x, y)$ plays the role of $b_t(x)$: it is the dynamical object that is discretized. A few-step rate model likewise chooses a path (or rate schedule), an approximation scheme, and a budget. The analogue of NFE is the number of generator or rate evaluations, or the number of accepted jumps, depending on the scheme. Nothing in the Gaussian ODE study proves a statement about CTMCs. The transferable discipline is: isolate a reduced ranking statistic, then test whether it actually orders endpoint error at fixed budget.

A13. References I

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